US012415393B2

(12) **United States Patent**
Gibbs et al.

(10) **Patent No.: US 12,415,393 B2**
(45) **Date of Patent: Sep. 16, 2025**

(54) **AMPHIBIAN**

(71) Applicant: **Gibbs Technologies Limited**, Nuneaton (GB)

(72) Inventors: **Alan Timothy Gibbs**, London (GB); **Simon James Longdill**, Nuneaton (GB); **Matthew Watson Houston**, Nuneaton (GB)

(73) Assignee: **Gibbs Technologies Limited**, Nuneaton (GB)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **16/940,560**

(22) Filed: **Jul. 28, 2020**

(65) **Prior Publication Data**

US 2021/0039461 A1 Feb. 11, 2021

Related U.S. Application Data

(63) Continuation of application No. 15/517,594, filed as application No. PCT/GB2015/052992 on Oct. 12, 2015.

(30) **Foreign Application Priority Data**

Oct. 10, 2014 (GB) 1418008
Jun. 8, 2015 (GB) 1509901

(51) **Int. Cl.**
B60F 3/00 (2006.01)

(52) **U.S. Cl.**
CPC ***B60F 3/0069*** (2013.01); ***B60F 3/0007*** (2013.01); ***B60F 3/003*** (2013.01); ***B60F 3/0038*** (2013.01)

(58) **Field of Classification Search**
CPC B60F 3/00; B60F 3/0007; B60F 3/003; B60F 3/0038; B60F 3/0061; B60F 3/0069; B60F 3/0084
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,269,349 A * 8/1966 Gehlen E01D 15/22 14/2.6
3,486,477 A 12/1969 Pender
(Continued)

FOREIGN PATENT DOCUMENTS

CN 1213482 C 8/2005
CN 100343456 C 10/2007
(Continued)

OTHER PUBLICATIONS

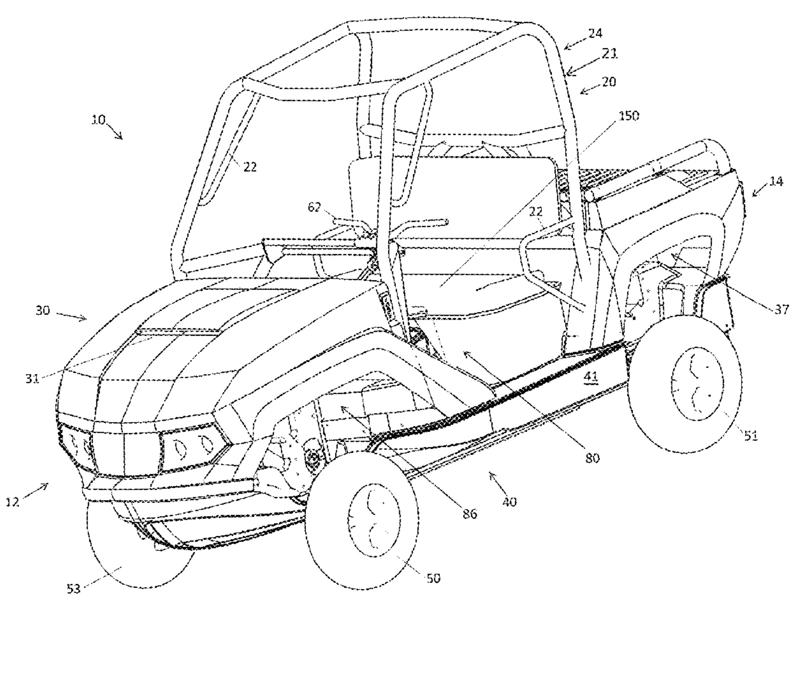
Chinese Patent Office; Search Report; Oct. 12, 2015; entire document.

Primary Examiner — Daniel V Venne
(74) *Attorney, Agent, or Firm* — Price Heneveld LLP

(57) **ABSTRACT**

An amphibian includes a planing hull, first and second sides, at least one retractable wheel, at least one prime mover, at least one steerable wheel which is connected to a steering control device operable by a driver to steer the amphibian, wherein the steering control device includes at least one of handlebars, a steering wheel, and steering levers, and is positioned to be grasped by an operator while operating the amphibian and wherein the steering control device is moveable between at least two distinct laterally spaced positions across a width of the amphibian between a center line and the respective sides of the planing hull.

22 Claims, 10 Drawing Sheets



(58) **Field of Classification Search**
USPC 440/12.5, 12.51, 12.52, 12.54, 12.57, 440/12.59, 12.66
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,765,368	A *	10/1973	Asbeck	B63C 13/00 440/12.54
3,898,949	A	8/1975	Kearsey	
3,903,831	A *	9/1975	Bartlett	B60F 3/00 114/344
3,933,112	A	1/1976	Veazey	
5,341,895	A	8/1994	Grinberg et al.	
5,417,177	A	5/1995	Taguchi	
5,590,617	A *	1/1997	Gere	B60F 3/0007 440/12.51
5,689,669	A	11/1997	Engler	
5,727,494	A	3/1998	Caserta	
5,832,862	A	11/1998	Hulten	
5,836,635	A	11/1998	Dorman	
RE36,901	E	10/2000	Roycroft et al.	
6,482,052	B1 *	11/2002	Giljam	B60F 3/0007 440/12.51
6,580,541	B1 *	6/2003	Yamashita	B60F 3/0007 398/104
6,666,735	B2	12/2003	Benoit	
7,934,963	B1	5/2011	Carambat	
8,221,174	B2 *	7/2012	March	B60F 3/0038 440/12.51
2002/0002939	A1 *	1/2002	Beauchesne	B60F 3/0007 114/382
2005/0283288	A1 *	12/2005	Howell	B62D 1/18 701/41
2007/0093148	A1	4/2007	Gibbs et al.	
2008/0227344	A1 *	9/2008	Gaither	B60F 3/0007 440/12.5
2009/0004932	A1 *	1/2009	Jeffery	B60F 3/0007 440/12.51
2012/0010818	A1	1/2012	Kalish	
2012/0108118	A1 *	5/2012	Longdill	B62K 9/00 440/12.52
2014/0130658	A1 *	5/2014	Gonzalez	F41H 7/042 89/36.08
2014/0288763	A1 *	9/2014	Bennett	B62D 11/18 701/31.6

FOREIGN PATENT DOCUMENTS

CN	101087699	A	12/2007
CN	103434560	A	12/2013
DE	2350971	A1	4/1975
EP	1581402	A	10/2005
FR	2385562	A1	10/1978
GB	2392415		3/2004
GB	2401829	A	11/2004
WO	9015747		12/1990
WO	9523074		8/1995
WO	2000074960	A1	12/2000
WO	2004020228	A1	3/2004

* cited by examiner

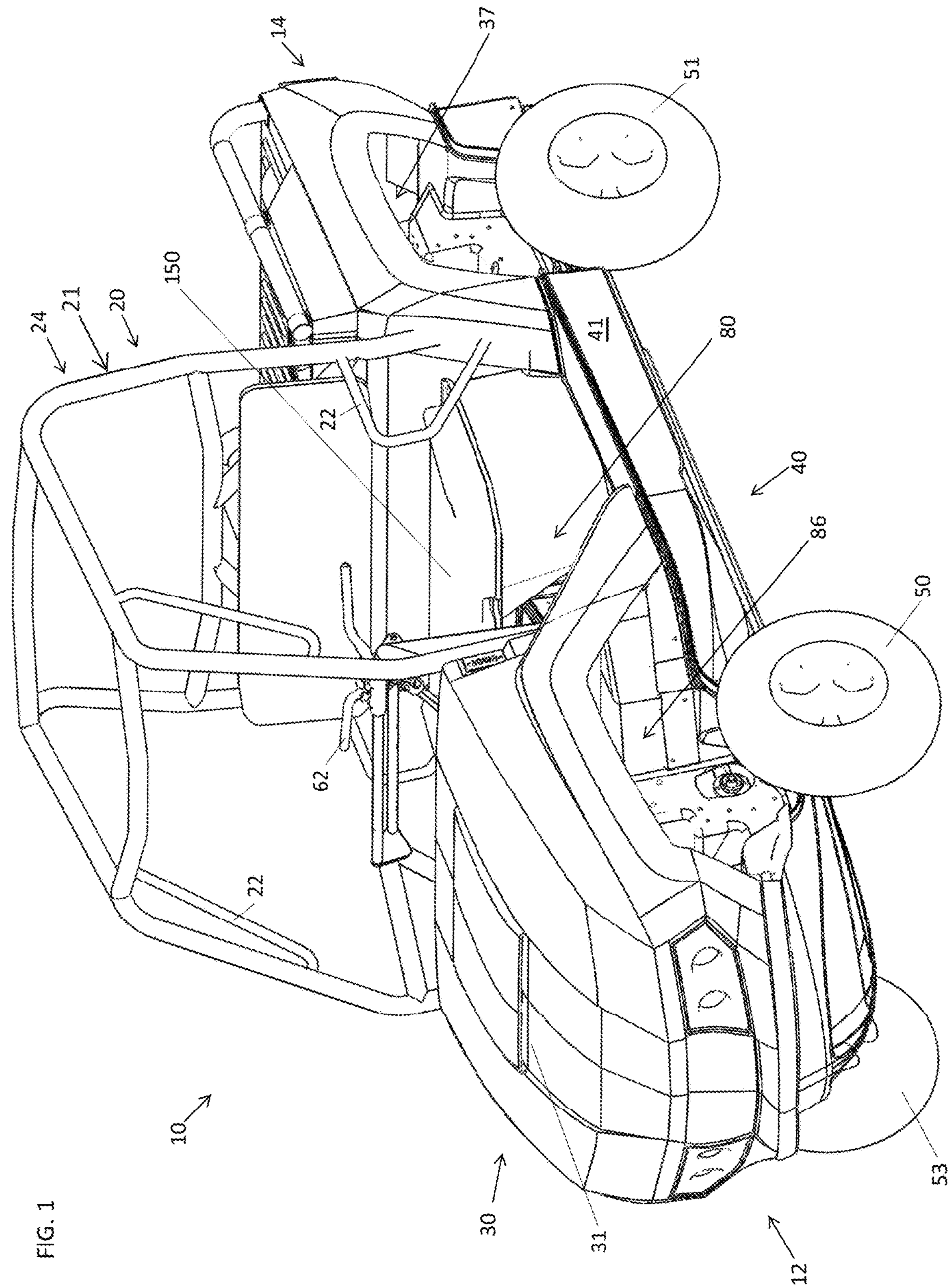


FIG. 1

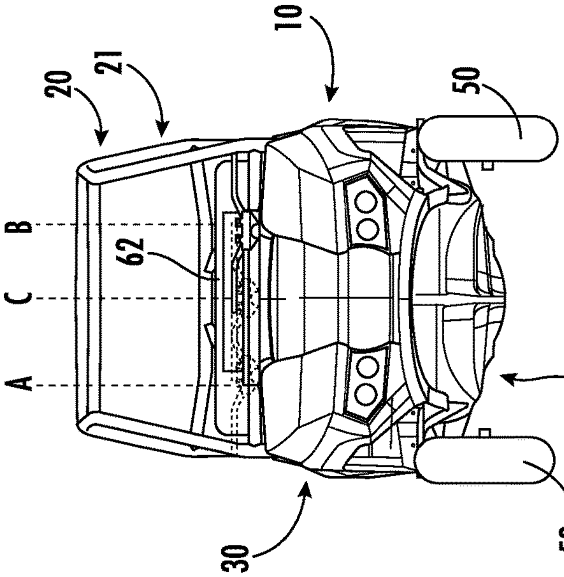


FIG. 3

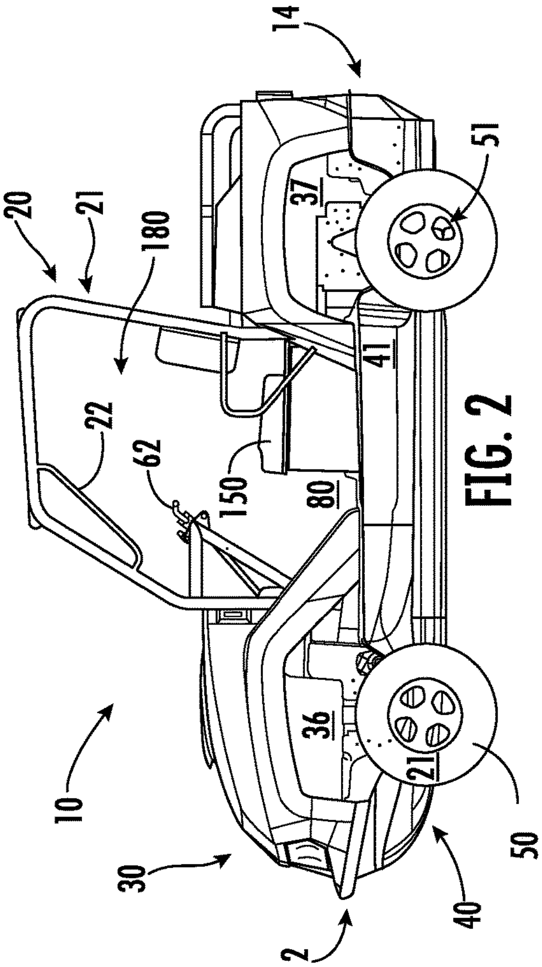


FIG. 2

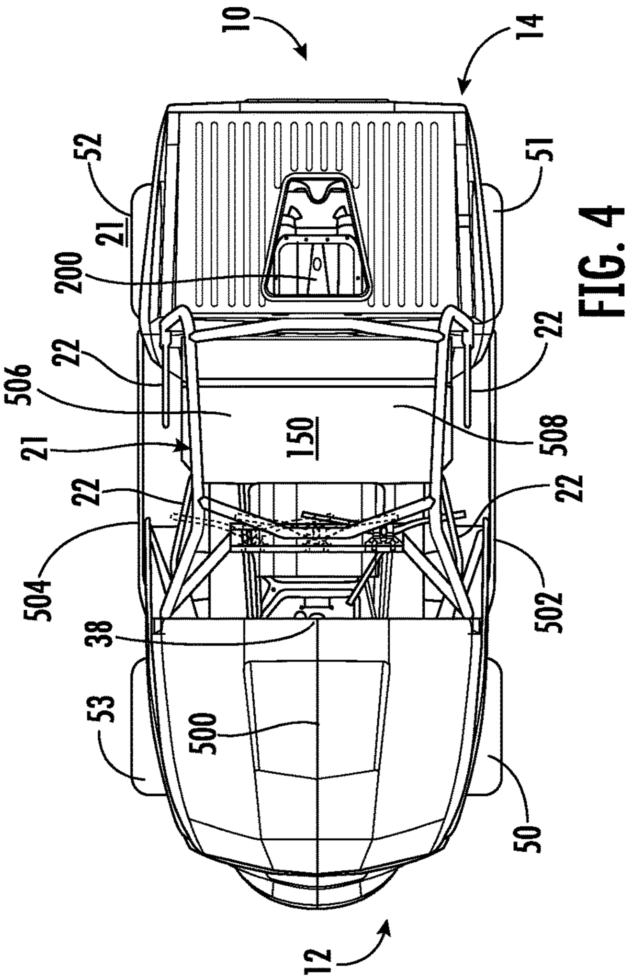


FIG. 4

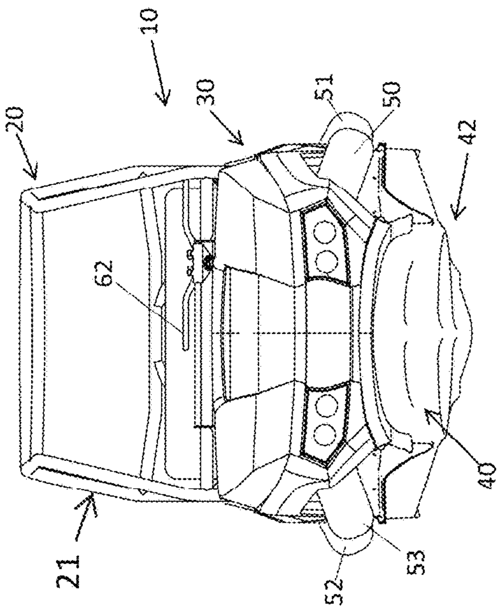


FIG. 6

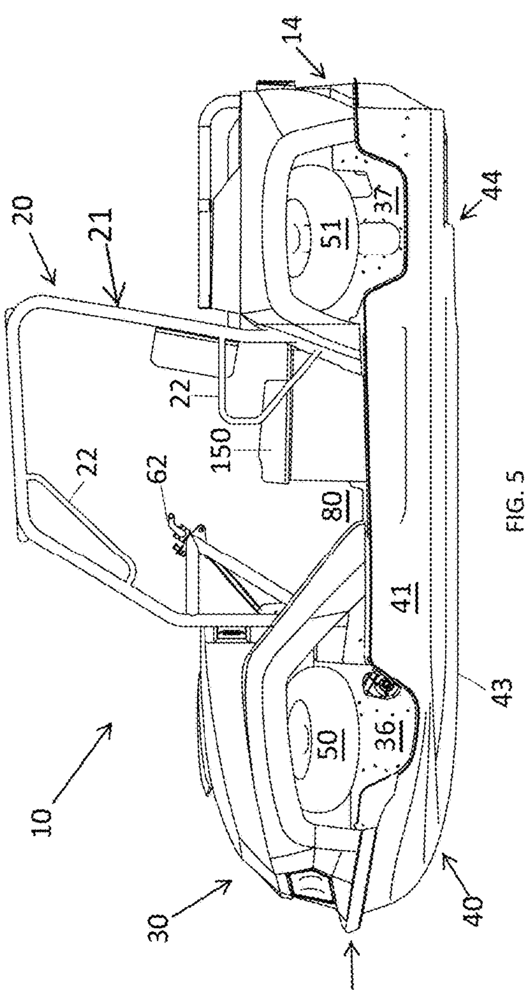


FIG. 5

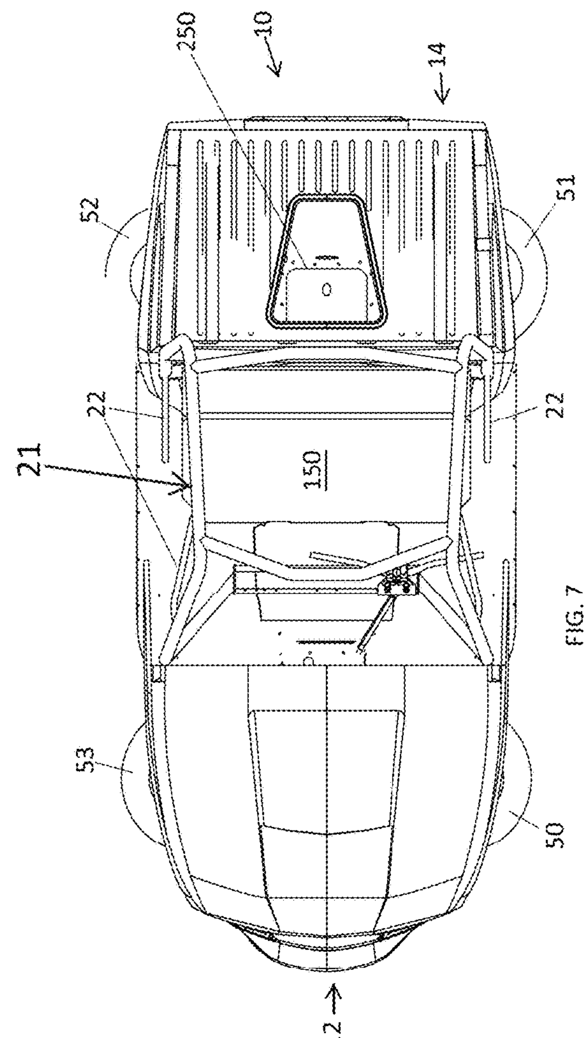


FIG. 7

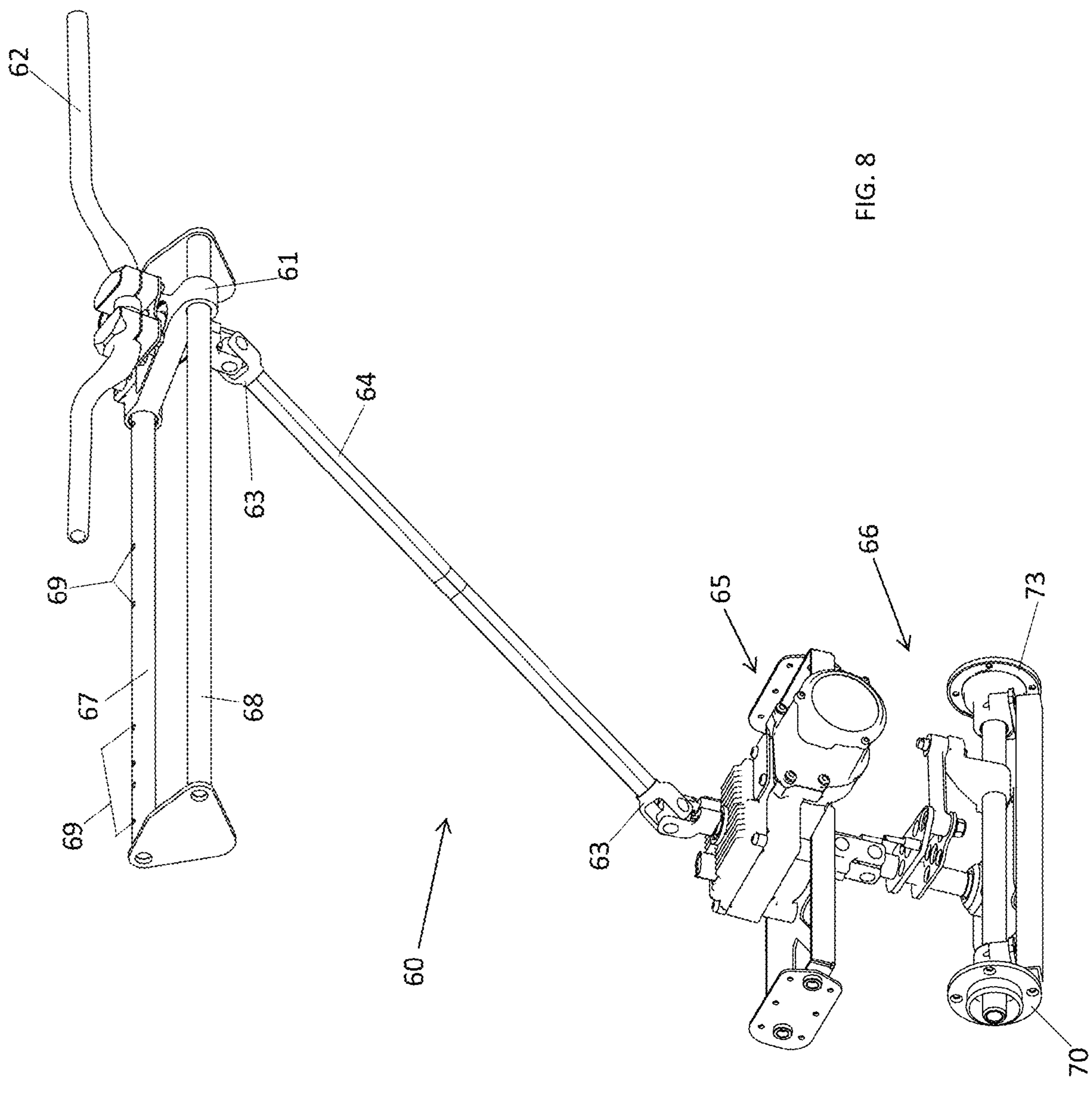


FIG. 8

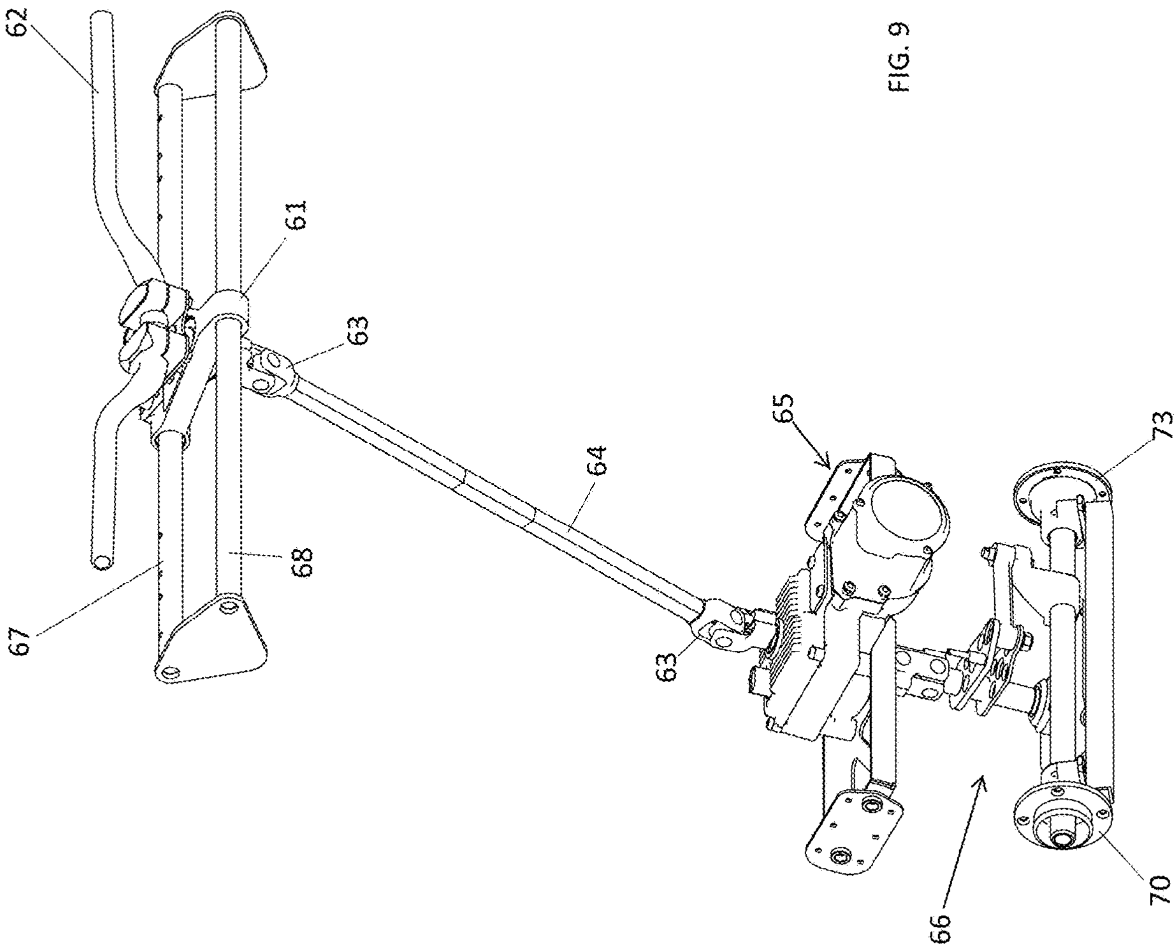


FIG. 9

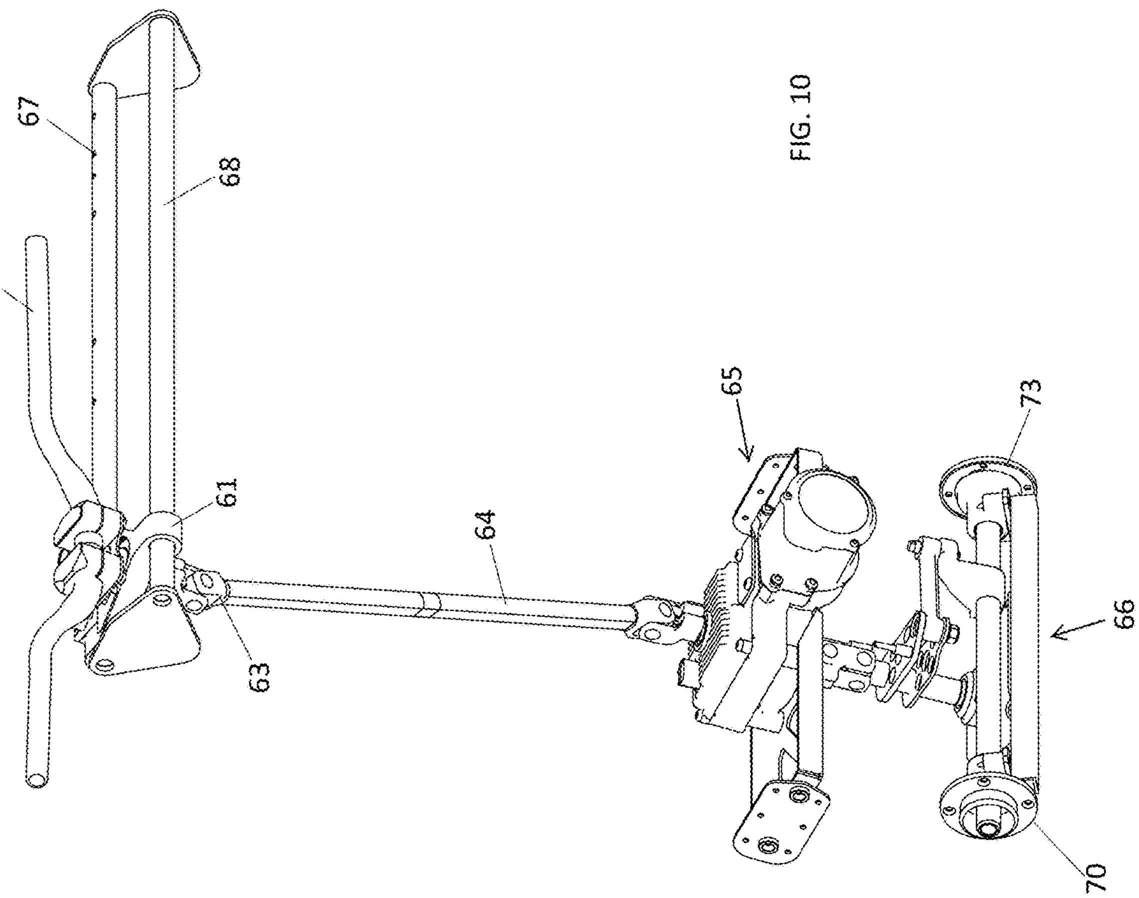


FIG. 10

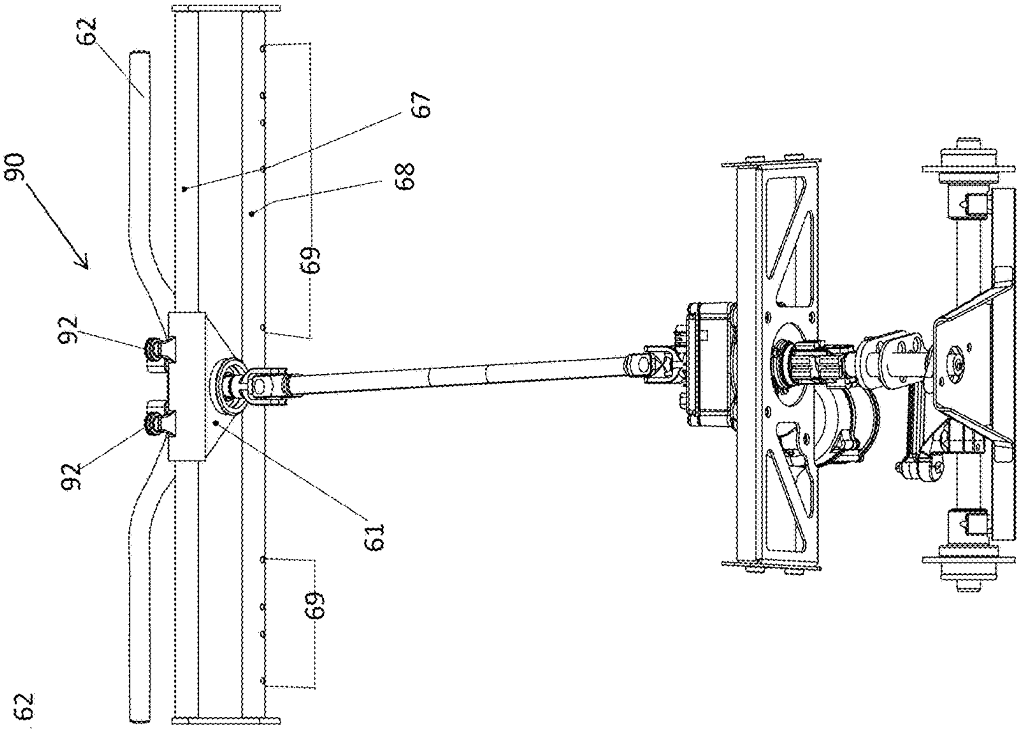


FIG. 12

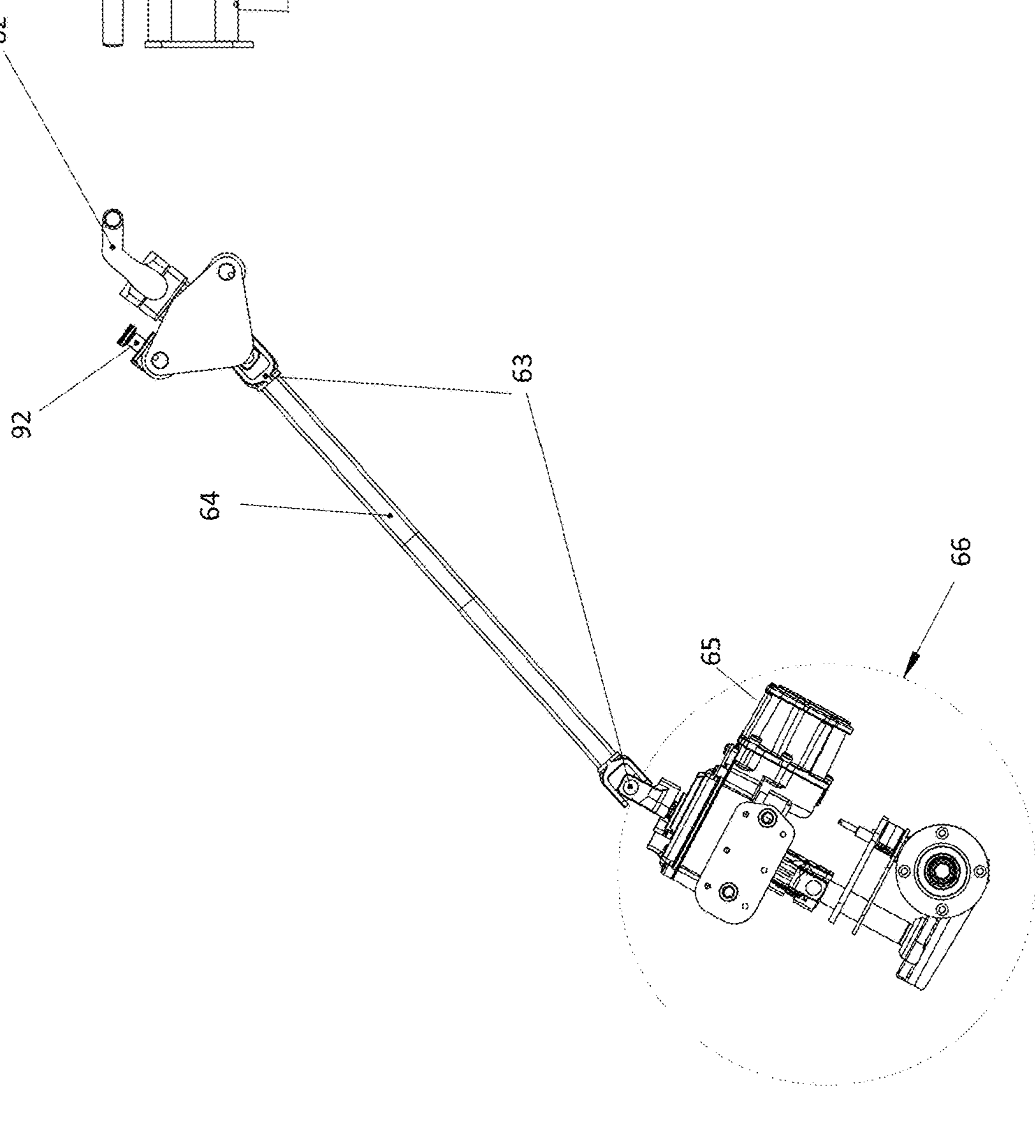


FIG. 11

FIG. 13

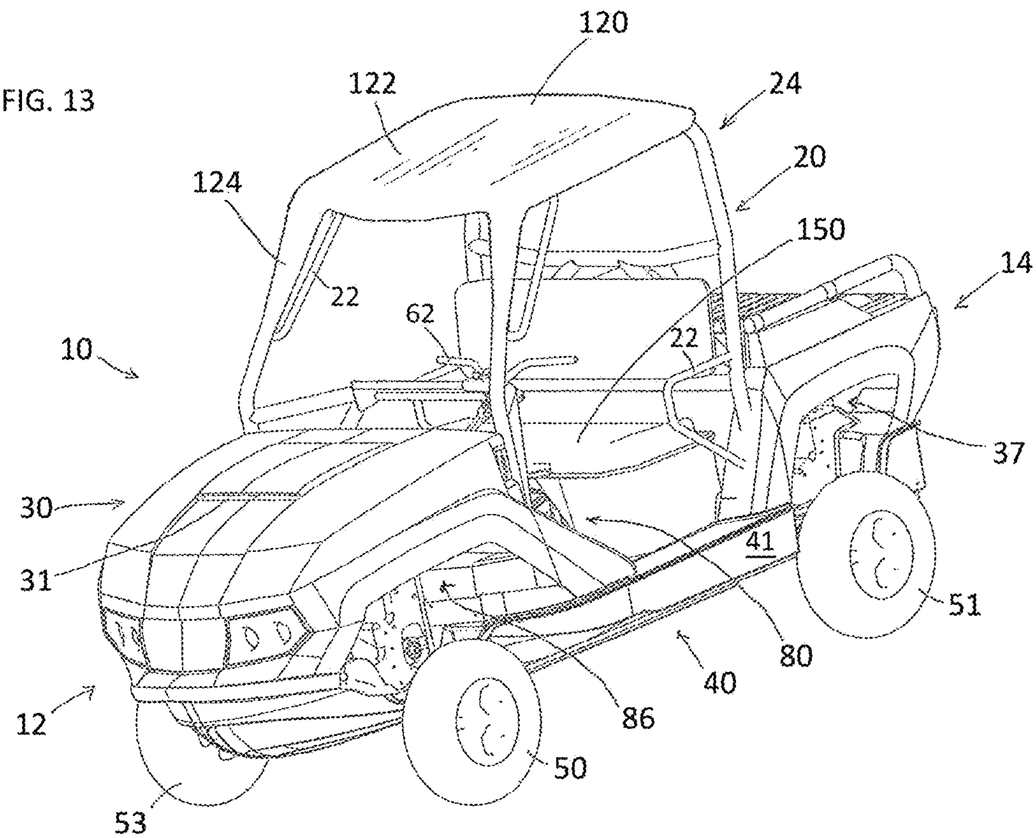


FIG. 14

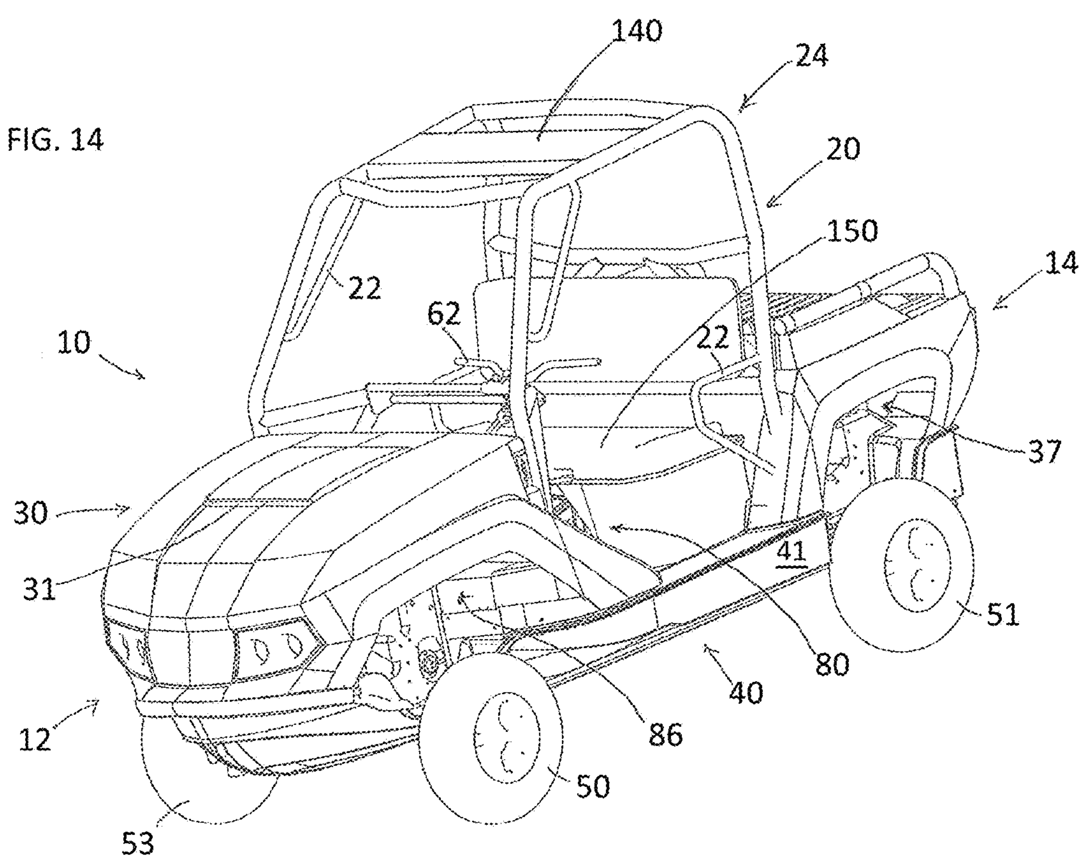
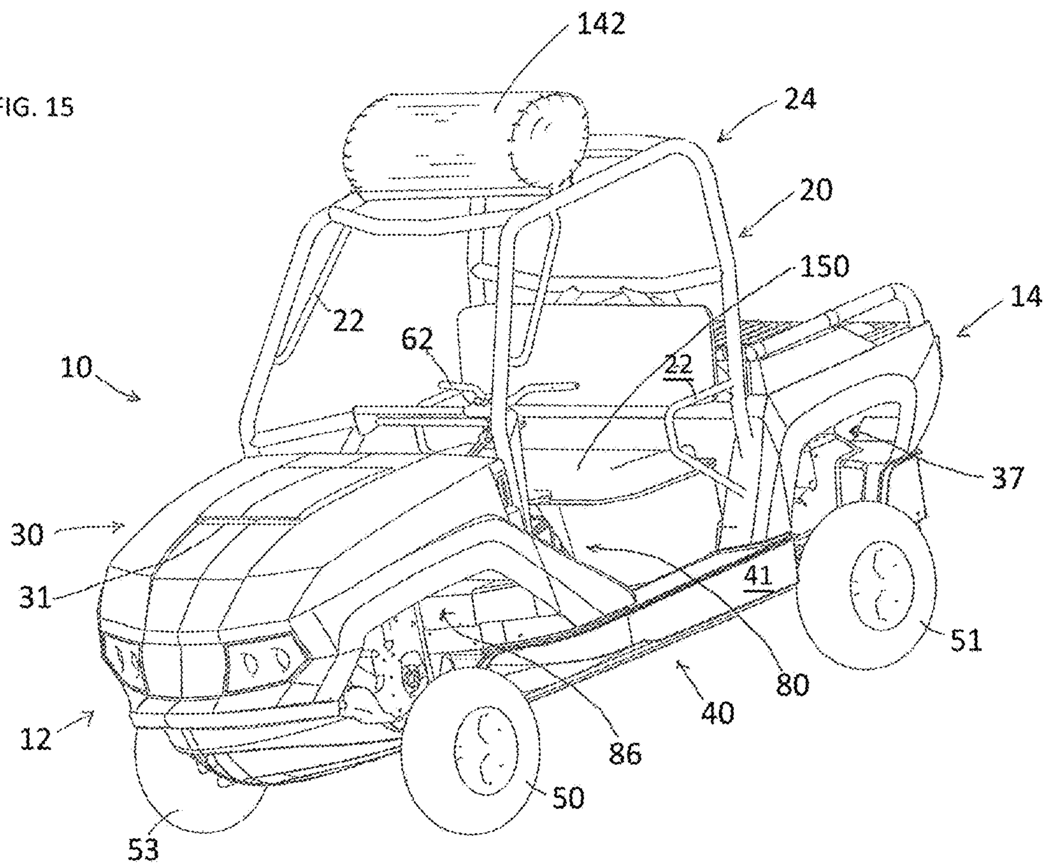


FIG. 15



AMPHIBIAN

CROSS-REFERENCE OF RELATED APPLICATIONS

The present application is a continuation of U.S. patent application Ser. No. 15/517,594, filed Apr. 7, 2017, entitled "AMPHIBIAN," which claims benefit to PCT Application No. PCT/GB2015/052992, filed Oct. 12, 2015, entitled "AMPHIBIAN," the entire disclosures of which are incorporated herein.

BACKGROUND OF THE INVENTION

The present invention relates to an amphibian and, in particular, to a high speed amphibian having what is typically referred to in land-going vehicles as utility task vehicle (UTV) capability. Preferably, the amphibian has seating arranged in a side-by-side manner.

Land-going UTVs and all terrain vehicles (ATVs) are well known in the art and have certain similarities. However, they are distinct vehicles intended for different uses and so adopt different configurations. An ATV is typically an off-road, sit-astride vehicle intended for use by a rider as a recreational vehicle, with capability for riding over rough terrain. A UTV is also intended for use over rough terrain, but this vehicle is driven rather than ridden and often referred to as a side-by-side vehicle because two or more people can sit in the driver area or cab next to each other on a bench seat(s) or bucket seat(s). In addition, a UTV is typically provided with a roll bar or roll cage arrangement. Further, a UTV is designed to carry or haul loads and so typically features a cargo area or truck-like bed specifically for this purpose.

Amphibians are now well known in the art. An amphibian is a vehicle which is able to move on both land and water. Historically, prior art amphibians either optimised operation in the land mode when used on land or, alternatively, optimised operation in the marine mode when used in water. The result was an amphibian having poor performance in one mode of operation or the other.

SUMMARY OF THE INVENTION

The present applicant has identified a need for an amphibian having UTV capability on land and which performs as a high speed watercraft on water.

In a first aspect, the present invention provides an amphibian operable on land and in/on water, comprising:

a planing hull;

at least one retractable wheel, the at least one retractable wheel being retractable from a protracted land engaging position when the amphibian is operated on land to a retracted position when the amphibian is operated in/on water;

at least one prime mover for providing power to drive, directly or indirectly, the amphibian on land and in/on water;

at least one steerable wheel which is, at least when the amphibian is operated on land, connected directly or indirectly to a steering control device operable by a driver to steer the amphibian; and wherein

the steering control device is moveable between at least two distinct laterally spaced positions across the width or beam of the amphibian.

In a second aspect, the present invention provides an amphibian operable on land and in/on water, comprising:

a planing hull;

at least one retractable wheel, the at least one retractable wheel being retractable from a protracted land engaging position when the amphibian is operated on land to a retracted position when the amphibian is operated in/on water;

at least one prime mover for providing power to drive, directly or indirectly, the amphibian on land and in/on water;

at least one steerable wheel which is, at least when the amphibian is operated on land, connected directly or indirectly to a steering control device operable by a driver to steer the amphibian; and

side-by-side seating.

In a third aspect, the present invention provides an amphibian operable on land and in/on water, comprising:

a planing hull;

at least one retractable wheel, the at least one retractable wheel being retractable from a protracted land engaging position when the amphibian is operated on land to a retracted position when the amphibian is operated in/on water;

at least one prime mover for providing power to drive, directly or indirectly, the amphibian on land and in/on water;

at least one steerable wheel which is, at least when the amphibian is operated on land, connected directly or indirectly to a steering control device operable by a driver to steer the amphibian; and

a roll over protection system.

In a fourth aspect, the present invention provides an amphibian operable on land and in/on water, comprising:

a planing hull;

at least one retractable wheel, the at least one retractable wheel being retractable from a protracted land engaging position when the amphibian is operated on land to a retracted position when the amphibian is operated in/on water;

at least one prime mover for providing power to drive, directly or indirectly, the amphibian on land and in/on water;

at least one steerable wheel which is, at least when the amphibian is operated on land, connected directly or indirectly to a steering control device operable by a driver to steer the amphibian; and

a cargo area or truck-like bed, preferably open.

In a fifth aspect, the present invention provides an amphibian operable on land and in/on water, comprising:

a planing hull;

at least one retractable wheel, the at least one retractable wheel being retractable from a protracted land engaging position when the amphibian is operated on land to a retracted position when the amphibian is operated in/on water;

at least one prime mover for providing power to drive, directly or indirectly, the amphibian on land and in/on water;

at least one steerable wheel which is, at least when the amphibian is operated on land, connected directly or indirectly to a steering control device operable by a driver to steer the amphibian; and

two seat belts and three seat belt buckles or buckle positions, or two seat belt buckles or buckle positions, and three seat belts.

In a sixth aspect, the present invention provides an amphibian operable on land and in/on water, comprising:

a planing hull;

at least one retractable wheel, the at least one retractable wheel being retractable from a protracted land engaging position when the amphibian is operated on land to a retracted position when the amphibian is operated in/on water;

at least one prime mover for providing power to drive, directly or indirectly, the amphibian on land and in/on water;

at least one steerable wheel which is, at least when the amphibian is operated on land, connected directly or indirectly to a steering control device operable by a driver to steer the amphibian; and

a footwell area having no transmission tunnel.

In a seventh aspect, the present invention provides an amphibian operable on land and in/on water, comprising:

a planing hull;

at least one retractable wheel, the at least one retractable wheel being retractable from a protracted land engaging position when the amphibian is operated on land to a retracted position when the amphibian is operated in/on water;

at least one prime mover for providing power to drive, directly or indirectly, the amphibian on land and in/on water;

at least one steerable wheel which is, at least when the amphibian is operated on land, connected directly or indirectly to a steering control device operable by a driver to steer the amphibian; and

a footwell and seating area configured so as not to impede or restrict driver and/or passenger entry via one side of the amphibian, egress from the other side of the amphibian, and through movement therebetween.

In an eighth aspect, the present invention provides an amphibian operable on land and in/on water, comprising:

a planing hull;

at least one retractable wheel, the at least one retractable wheel being retractable from a protracted land engaging position when the amphibian is operated on land to a retracted position when the amphibian is operated in/on water;

at least one prime mover for providing power to drive, directly or indirectly, the amphibian on land and in/on water;

at least one steerable wheel which is, at least when the amphibian is operated on land, connected directly or indirectly to a steering control device operable by a driver to steer the amphibian; and

a positive buoyancy system and/or deployable righting system.

In a ninth aspect, the present invention provides an amphibian operable on land and in/on water, comprising:

a planing hull;

at least one retractable wheel, the at least one retractable wheel being retractable from a protracted land engaging position when the amphibian is operated on land to a retracted position when the amphibian is operated in/on water;

at least one prime mover for providing power to drive, directly or indirectly, the amphibian on land and in/on water;

at least one steerable wheel which is, at least when the amphibian is operated on land, connected directly or indirectly to a steering control device operable by a driver to steer the amphibian; and

an interlock or safety system which prevents operation/activation of at least one driver or automatically controlled parameter of the amphibian unless sensing/determining a first condition has been satisfied.

These and other features, advantages and objects of the present invention will be further understood and appreciated by those skilled in the art by reference to the following specification, claims and appended drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

Preferred embodiments of the present invention will now be described by way of example only with reference to the accompanying drawings, in which;

FIG. **1** is a schematic perspective view from the front and above of an amphibian according to the present invention with its wheels shown in a protracted land engaging position (certain parts omitted for clarity);

FIG. **2** is a schematic side elevation view of the amphibian of FIG. **1** with its wheels shown in a protracted land engaging position;

FIG. **3** is a schematic front elevation view of the amphibian of FIG. **1** with its wheels shown in a protracted land engaging position;

FIG. **4** is a schematic top plan view of the amphibian of FIG. **1** with its wheels shown in a protracted land engaging position;

FIG. **5** is a schematic side elevation view of the amphibian of FIG. **1** with its wheels shown in a retracted position;

FIG. **6** is a schematic front elevation view of the amphibian of FIG. **1** with its wheels shown in a retracted position;

FIG. **7** is a schematic top plan view of the amphibian of FIG. **1** with its wheels shown in a retracted position;

FIG. **8** is a schematic perspective view from behind and to the side of a steering assembly according to the present invention with a steering control device shown in a right hand position, spaced laterally from left hand and centre positions;

FIG. **9** is a schematic perspective view from behind and to the side of the steering assembly of FIG. **8** with the steering control device shown in a central position, spaced laterally from left hand and right hand positions;

FIG. **10** is a schematic perspective view from behind and to the side of the steering assembly of FIG. **8** with the steering control device shown in a left hand position, spaced laterally from right hand and centre positions;

FIG. **11** is a schematic side elevation view of the steering assembly of FIG. **8** with certain parts omitted for clarity;

FIG. **12** is a schematic front elevation view of the steering assembly of FIG. **8** with the steering control device shown being moved to the right from a central position;

FIG. **13** is a schematic perspective view from the front and above of the amphibian of FIG. **1** provided with an optional positive buoyancy system;

FIG. **14** is a schematic perspective view from the front and above of the amphibian of FIG. **1** provided with an optional deployable righting system; and

FIG. **15** is a schematic perspective view from the front and above of the amphibian of FIG. **14** with the deployable righting system deployed.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENT

The amphibian **10** has four road wheels **50**, **51**, **52**, **53** which are connected to the remainder of the amphibian **10** by a wheel suspension and retraction system (omitted for clarity) for moving the wheels **50**, **51**, **52**, **53** between a protracted land engaging position for use on land in a land mode of the amphibian **10** and a retracted position for use in/on water in a marine mode of the amphibian **10**. The front wheels **50** and **53** are steerable by a driver of the amphibian **10** via a steering assembly **60**, with a steering control device in the form of handlebars **62** in the preferred embodiment shown. Alternatively, the steering control device could take the form of a steering wheel, steering levers, or any other suitable steering control device capable of conveying a steering input from the driver. The rear wheels **51**, **52** are driven to propel the amphibian **10** on land in the preferred embodiment shown, although as an alternative front and/or four wheel drive may be employed. Further, as an alternative, any one or more of the wheels may be driven in use of the amphibian **10**. One or more marine propulsion devices **300** propel the amphibian **10** in/on water, and this/these may take the form of a jet drive(s) and/or propeller(s), for example, where the marine propulsion devices **300** may be conventional. Power is provided by way of one or more prime movers **200**, such as an internal combustion engine, electric or hydraulic motor, and may be delivered to the wheel(s) and/or marine propulsion device(s) **300** via a transmission directly, or indirectly via a gearbox or other speed change device such as a speed change transmission **250** (e.g. continuously variable transmission) used to vary the speed of the output from the prime mover(s) **200** relative to the input drive to the one or more driven wheels and/or marine propulsion device(s) **300**, where the speed change transmission may be conventional.

The structure of the amphibian **10** comprises an upper body **30**, a roll over protection system (ROPS) **20** in the form of a roll cage, and a hull **40**. The body **30** and the roll cage of the ROPS **20** are separately connected to (but demountable from, as a whole or in parts) the hull **40**. This permits ease of access to internal components of amphibian **10** for servicing, etc. A cargo area or truck-like bed is provided, preferably open, and preferably behind the roll over protection system (ROPS). Alternatively, the roll over protection system (ROPS) may extend to at least partially afford protection to the cargo area or truck-like bed. In an alternative embodiment, the cargo area or truck-like bed is provided close to the water line so as to ease loading and unloading, e.g. for launch and/or recovery of people in a rescue situation. As a further alternative, the cargo area or truck-like bed may be moveable between a first position above the water line and a second position provided close to the water line for ease loading and unloading.

The roll cage of the ROPS **20** forms an integral part of the entire structure of the amphibian **10**. It is a structural component and not merely cladding. Typically it takes the form of one or more elongate tubular or angled section structural members, one or more of which are directly connected to the composite structure of the hull **40** (e.g. glass fibres or carbon fibres set in resin) and not via an intermediate frame or chassis. As such, overall weight is reduced yet stiffness and strength improved. Handles and/or grab rails **22** are optionally provided. In addition, or alternatively, optional webbing 'doors' and/or restraint bars (not shown) can be provided where doors in a traditional vehicle are provided. The roll cage of the ROPS **20** provides an ideal structure for mounting or location of these optional handles and/or grab rails **22** and webbing 'doors' and/or restraint bars. Where localised areas of strength are required to help form or enhance the integrity of the connection of the roll cage **20** of the ROPS with the body **30** and/or hull **40**, extra layers or mats of fibres and/or or other (e.g. metal) components may be laid down in the body **30** and/or hull **40** during manufacture. The body **30** and/or hull **40** themselves can be formed with localised reinforced areas as necessary, and provide a complete support structure for components making up the amphibian **10**. In the preferred form of roll cage of the ROPS **20** shown, a cab area **180** is defined and is where a driver and passengers of the amphibian **10** are seated in use. An upper frame portion **24** of the roll cage **20** of the ROPS can be seen protrude above the upper surfaces of the body **30**, providing over head, front, rear and side protection for the driver and passengers of the amphibian **10**. Furthermore, the roll cage of the ROPS **20** provides an ideal structure for mounting or location of a towing hook and/or rope, e.g. for use in wakeboarding where the added height provides for more lift when conducting jumps and/or aerial tricks. Yet further, the roll cage of the ROPS **20** provides an ideal structure for i) storage of items, ii) mounts for locating or storing items, or iii) mounting or location of storage bins/containers, e.g. for wakeboards, waterskis, fishing rods or other sporting apparatus, and/or tools.

A side-by-side seating configuration is provided in the form of a bench seat **150**. The bench seat **150** extends laterally across (transversely) the width (beam) of the amphibian **10**, including a first seating position **506** and a second seating position **508**. This provides for at least three distinct laterally spaced offset driving positions of the amphibian **10**, i) right hand drive or first position A, ii) centre drive or third position C and iii) left hand drive or second position B. Optional seatbelts and seatbelt buckles, or harnesses, may be provided, as necessary or required by local legislation. Again, the roll cage of the ROPS **20** provides an ideal structure for mounting or location of these seatbelts and seatbelt buckles, or harnesses. Where a three seat configuration is adopted as described above, it has been determined by the applicant that a particularly effective arrangement is to have either two seat belts and three seat belt buckles or buckle positions, or two seat belt buckles or buckle positions, and three seat belts. In the former case, by arranging the three seatbelt buckles/positions spaced laterally across the amphibian and/or bench seat **150** with two seat belts, it has been found that this can provide an optimised solution which does not restrict entry to, or egress from, the amphibian **10**. Similarly, in the latter case, by arranging the two seatbelt buckles/positions spaced laterally at, for example, substantially the ⅓ and ⅔ positions across the amphibian and/or bench seat **150** with three seat belts, it has been found that this can provide an optimised solution which does not restrict entry to, or egress from, the amphibian **10**. As will be described further below, the steering assembly in fact provides for a multitude of driving positions across the full width of the amphibian, providing a flexibility which increases the functionality of the amphibian **10**. As an alternative to a bench seat **150**, one or more bucket seats may instead be provided. Indeed, this provides further flexibility, as one or more bucket seats and/or bench seats may be modular in nature and utilised and positioned as necessary and accordingly across the width (beam) of the amphibian **10**, so as to provide for various driving and passenger positions and configurations, Indeed, multiple rows of seating may be provided, even using a mix of seating types and arrangements. Where bucket seats are used, these can be mounted so as to permit a swivel action to enable a driver and/or passenger(s) to, for example, fish out of a/each side of the amphibian **10** while seated. A complimentary footwell area **80** also extends laterally across (transversely) the width (beam) of the amphibian **10**, so as to accommodate all possible driving positions. The footwell area **80** is optionally provided with means to bail automatically any water shipped in use of the amphibian **10**. One such example of a means to bail automatically is the provision of a cambered floor, or cambered floor portions (e.g. arranged with a central peak), in the footwell area **80** to direct and channel water away and out of the amphibian **10**.

It will be appreciated from the foregoing that the seating and footwell area is configured so as not to impede or restrict the driver and/or passengers but instead to facilitate entry via either side of the amphibian **10**, egress from either side of the amphibian **10**, and through movement either way therebetween. Indeed, to further assist in this regard, the function of traditional foot pedals can be integrated with the steering assembly or elsewhere in the amphibian **10**, as described further below, thus avoiding the need altogether for any foot pedals in the seating and footwell area. Alternatively, the function of traditional foot pedals may be provided by one or more sets of foot pedals, situated as necessary (e.g. at one or more driving positions), in the footwell area or otherwise. Indeed, these foot pedals may be retractable or foldable to clear the footwell or other area, and/or be slidable across the amphibian in a similar manner to that of the steering control device.

Front and rear wheel arches **36**, **37** are provided on either side of the amphibian **10** so as to at least partially contain a wheel suspension and/or retraction assembly which is retracted when the amphibian **10** is operating in/on water in the marine mode.

Air inlet opening **31** provides an entry for cooling air (ram and/or fan-assisted) for use by the cooling systems of the amphibian **10**. Air entrained via inlet **31** is eventually exhausted via an air outlet (not shown). Between air inlet **31** and air outlet, a dorade system is installed to prevent the ingress of water.

An instrument panel **38** is provided in the cab area **180** to convey relevant parameters (e.g. operating, control and/or status) of the amphibian **10** to the driver. Additionally, side and/or rear view mirrors (not shown) may be provided as necessary as a visual aid to the driver. Furthermore, land and marine navigation lights may also be provided and operate in accordance with the local legislative requirements.

The hull **40** can be seen extending along a longitudinal axis or longitudinally-extending center line **500** of the amphibian **10** from the front bow section **12** to the rear stern section **14**, and across a transverse axis of the amphibian **10**. Indeed, the hull extends across the width (beam) of the amphibian **10** between the front and rear wheels at hull portion **41** and between a first side **502** and a second side **504**. Hull portion **41** provides stability when the amphibian **10** is operated at high speed on water in the marine mode because it provides displacement spaced laterally from the centre line **500** of the amphibian **10**. As such, when cornering sharply, for example, an increase in righting force is experienced as the angle of lean increases.

The hull **40** comprises a V formation **42** with a keel section **43** running from the bow **12** of the amphibious vehicle to a point where the keel splits to incorporate a water intake area **44** for a jet drive marine propulsion unit **300**. Consequently, the wheels **50**, **51**, **52**, **53** must be protracted and retracted through a greater height in order to provide for good ground clearance of the hull **40** when the amphibian **10** is used on land in land mode, yet provide for the wheels **50**, **51**, **52**, **53** to be clear of the waterline when the amphibian **10** is used in/on water in marine use. However, this strategy does provide for an amphibian having UTV capability both on land and on water. Even with the small footprint of the hull **40** of the amphibian **10**, the hull **40** is capable of propelling the amphibian **10** up onto the plane with little difficulty in fast time periods. Surprisingly, in/on-water performance of the amphibian is not compromised and adequate ground clearance is available even for off-road use.

Next, with reference to FIGS. **8** to **12**, the steering assembly **60** can be seen. In FIG. **8** it can be seen that the handlebars **62** are connected by a steering column **64** to a steering mechanism **66** capable of steering the hubs **70**, **73**. Optional power assisted steering **65** is provided. The hubs **70**, **73** connect to front wheels **50**, **53** via a suspension upright (omitted for clarity). At each end of the steering column **64** is a universal joint **63** to provide for rotation and articulation. The steering column **64** comprises a splined connection to provide for both the rotation and extension necessary. The handlebars **62** are mounted via a handlebar bearing housing **61** to upper **67** and lower **68** handlebar guide rails which guide lateral translation of the handlebars **62**. Together, these features facilitate movement of the handlebars laterally across (transversely) the width (beam) of the amphibian **10**. This provides for at least three distinct driving positions of the amphibian **10**, i) right hand drive or first position A located between the centre line **500** and the second side **504**, ii) centre drive or third position C located along the centre line **500** and iii) left hand drive or second position B located between the first side **502** and the centre line **500**.

An interlock safety device **90** comprises one or more biased locking pins **92** mounted via a handlebar bearing housing **61**, each locking pin **92** having a projection received in one of a series of guide holes **69** provided in either one or both of upper **67** and lower **68** handlebar guide rails. Proximity sensors and/or a “dead man” key can be utilised in addition to prevent starting of, or to ‘kill’, the prime mover **200** in the event that the handlebars **62** are not locked in place. To reposition the handlebars **62**, a driver can pull on the one or more biased locking pins **92** to remove the projection(s) from the respective guide hole or holes **69**, and then translate the handlebars **62** to the desired position laterally along the guide rails **67**, **68**, whereupon the one or more biased locking pins **92** can be released and the projection(s) engaged into the respective guide hole or holes **69** corresponding to the new steering position. Alternatively, a friction lever arrangement could be utilised to provide for unlimited lateral arrangement. Furthermore, height adjustment may also be provided, e.g. by rotation of the steering assembly **60** about a transverse axis of the amphibian **10**. While the steering control device described takes the form of handlebars **62**, it could alternatively take the form of a steering wheel, steering levers, or any other suitable steering control device capable of conveying a steering input from the driver, as described above. However, the applicant has found that the use of handlebars has particular advantage in terms of optimising the full extent of steering input angle required for both land and marine mode steering (a steering wheel typically requires more than one full rotation from lock to lock, whereas handlebars require only a fraction of a full rotation).

Yet further important benefits of the steering assembly disclosed herein are that i) it enables throttle and braking functions to be easily translated together with the steering input device without the need or complexity of having to move foot control pedals, and ii) there is no steering disconnect required between land and marine mode changes. Both can remain connected throughout use in both the land and the marine modes, reducing complexity (familiarisation/intuitiveness of use by the driver in both modes), and increasing safety (by avoiding the need for the disconnect/reconnect of separate systems).

The roll over protection system (ROPS) **20** is configured to serve multiple different functions in the amphibian **20** according to the present invention. Not only does it serve as a structural component, with its roll cage providing the important function of protecting occupants in a roll over situation on land, but it can also be used to incorporate systems to prevent inversion in a capsize situation on water, and/or to provide for proactive righting of the amphibian **10** in an inversion or capsize situation. By way of example, and with reference to FIGS. **13** to **15** in particular, the upper frame **24** of the roll cage of the ROPS **20** can be provided with a positive buoyancy system **120** and/or a deployable righting system **140**. The positive buoyancy system **120** is illustrated in FIG. **13** and takes the form of one or more low density (relative to water) closed cellular or enclosed volume structures **122**, **124** mounted on or to the upper frame **24**. This/these closed cellular or enclosed volume structures **122**, **124** provide positive buoyancy in a capsize situation on water, displacing volumes of water with a lower density medium to provide for positive buoyancy. This reduces the likelihood of an inversion of the amphibian **10**. Alternatively, or to complement the positive buoyancy system **120**, a deployable righting system **140** may be employed, as illustrated in FIGS. **14** and **15**. The deployable righting system **140** takes the form of an inflatable structure **142**, inflation source and control system with sensors. The sensors can be configured to detect roll angle of the amphibian **10** relative to a datum (e.g. fully upright) position, and the control system can be set to trigger inflation of the inflatable structure **142** by the inflation source when a selected threshold is reached or passed. Suitable safeguards can be implemented to avoid an unnecessary trigger, such as sensing also that the amphibian is in the marine mode and/or that, for example, a water sensor mounted on the upper frame **24** has been submerged in water. Upon inflation of the inflatable structure **142**, a significant righting force (given its position, i.e. distance up on the upper frame **24**) is realised which flips the amphibian **10** upright. Furthermore, the positive buoyancy system **120** and/or deployable righting system **140** can also be configured and employed in land mode, serving to protect the driver and passengers from impacts by cushioning (not dissimilar to an airbag function in a land-only going vehicle).

Preferably, spring and damper assemblies are provided for the wheels **50**, **51**, **52**, **53**. Hydraulic (or other) actuators can be provided to allow the wheels to be retracted from their protracted positions shown in FIGS. **1** to **4** to their retracted positions shown in FIGS. **5** to **7**. The hydraulic actuators are powered by hydraulic fluid supplies from a pump (not shown) powered by the prime mover **200**. Each wheel suspension assembly comprises a suspension upright member connected to the outboard ends of upper and lower suspension arms, and arranged for retraction between protracted and retracted positions. The wheels **50**, **51**, **52** and **53** are each mounted to a hub rotatably engaged with and/or carried on the suspension upright member. The hydraulic actuators each have a piston rod which acts retract and protract the suspension assemblies about an axis running fore and aft longitudinally along the amphibian. Other actuators (e.g. electric rotary or linear, or pneumatic) may be employed.

Cooling is provided via a radiator located at the front of (or otherwise) the amphibian **10** acted on by the air entering via air inlet opening **31** and which can cool the prime mover **200**, at least in use of the amphibian **10** on land. Cooling may additionally be provided by a water/water heat exchanger in use of the amphibian **10** in/on water, with water being drawn from outside the amphibian **10**. In addition, exhaust cooling may be employed using the same heat exchangers/system, or by using separate dedicated heat exchangers of the same or similar form.

Whilst wheels have been described throughout as the land propulsion means, track drives or individual track drives (i.e. to replace a single wheel) may be used as an alternative or in combination with wheels. As such a reference to wheel or wheels in the description and claims is to be construed as including a track drive or track drives.

Each feature disclosed in this specification (including the accompanying claims and drawings) may be replaced by alternative features serving the same, equivalent or similar purpose, unless expressly stated otherwise. Thus, unless expressly stated otherwise, each feature disclosed is one example only of a generic series of equivalent or similar features. In addition, all of the features disclosed in this specification (including the accompanying claims and drawings), and/or all of the steps of a method or process, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive. Accordingly, while different embodiments of the present invention have been described above, any one or more or all of the features described, illustrated and/or claimed in the appended claims may be used in isolation or in various combinations in any embodiment. As such, any one or more feature may be removed, substituted and/or added to any of the feature combinations described, illustrated and/or claimed. For the avoidance of doubt, any one or more of the features of any embodiment may be combined and/or used separately in a different embodiment with any other feature or features from any of the embodiments.

Whereas the present invention has been described in relation to what is presently considered to be the most practical and preferred embodiments, it is to be understood that the invention is not limited to the disclosed arrangements but rather is intended to cover various modifications and equivalent constructions included within the scope of the appended claims.

The invention claimed is:

1. An amphibian operable on land and on water, comprising:

a planing hull having a longitudinally-extending center line, a first side and a second side opposite the first side;

at least one retractable wheel, the at least one retractable wheel being retractable from a protracted land engaging position when the amphibian is operated on land to a retracted position when the amphibian is operated on water;

at least one prime mover for providing power to drive, directly or indirectly, the amphibian on either one or both of land and water;

at least one steerable wheel mechanically connected directly or indirectly to a steering control device operable by a driver to steer the amphibian, wherein the at least one steerable wheel and the steering control device operable by the driver remain connected throughout use in both the land and the marine modes such that there is no steering disconnect required during a change between land and marine modes, and wherein the steering control device includes at least one of handlebars, a steering wheel, and steering levers, and wherein the steering control device is positioned to be grasped by the driver while operating the amphibian; and

wherein the steering control device operable by the driver is moveable between at least two distinct laterally spaced positions across a width or beam of the amphibian including between a first position and a third position or between a second position and the third position, wherein the first position is located between the longitudinally-extending center line and the first side of the planing hull such that a majority of the steering control device is located between the longitudinally-extending center line and the first side of the planing hull, the second position is located between the longitudinally-extending center line and the second side of the planing hull such that a majority of the steering control device is located between the longitudinally-extending center line and the second side of the planing hull, and the third position is located centrally of the first and second sides of the planing hull such that at least a part of the steering control device is located along the longitudinally-extending center line.

2. The amphibian as claimed in claim **1**, wherein the amphibian is a utility task vehicle (UTV).

3. The amphibian as claimed in claim **1**, further comprising:

side-by-side seating.

4. The amphibian as claimed in claim **3**, wherein the side-by-side seating comprises one or more bench seats.

5. The amphibian as claimed in claim **1**, further comprising:

a roll over protection system.

6. The amphibian as claimed in claim **1**, further comprising:

either one or both of a positive buoyancy system and a deployable righting system.

7. The amphibian as claimed in claim **1**, further comprising:

a cargo area.

8. The amphibian as claimed in claim **7**, wherein the cargo area is positioned rearward of a crew or cab area.

9. The amphibian as claimed in claim **1**, further comprising:

a footwell area having no transmission tunnel.

10. The amphibian as claimed in claim **1**, wherein the footwell area does not comprise foot pedals.

11. The amphibian as claimed in claim **1**, further comprising:

a roll cage.

12. The amphibian as claimed in claim **1**, further comprising:

at least one marine propulsion device.

13. The amphibian as claimed in claim **1**, wherein the planing hull comprises a V formation.

14. The amphibian as claimed in claim **1**, further comprising:

a speed change transmission operably coupling the prime mover and one or both of i) at least one driven wheel or wheels and ii) at least one marine propulsion device.

15. The amphibian as claimed in claim **1**, wherein the at least one retractable wheel is pivoted through 15° or more during retraction.

16. The amphibian as claimed in claim **1**, wherein the at least one retractable wheel is retracted above a lowest point of the planing hull for use on water in marine mode, and is protracted below a lowest point of the planing hull for use on land in land mode.

17. The amphibian as claimed in claim **1**, wherein when the amphibian is operated in a land mode one or more of the at least one retractable wheel may be driven.

18. The amphibian of claim **1**, wherein the steering control device is moveable between the first position and the third position and between the second position and the third position.

19. The amphibian of claim **1**, wherein the handlebars are mounted to upper and lower handlebar guide rails by a bearing housing, where the handlebar guide rails guide lateral translation of the handlebars.

20. An amphibian operable on land and on water, comprising:

a planing hull;

at least one retractable wheel, the at least one retractable wheel being retractable from a protracted land engaging position when the amphibian is operated on land to a retracted position when the amphibian is operated on water;

at least one prime mover for providing power to drive, directly or indirectly, the amphibian on either one or both of land and water;

a seating arrangement having a first seating position for a driver of the amphibian, a second seating position for the driver of the amphibian that is entirely laterally offset from the first seating position, and a third seating position for the operator of the amphibian between the first and second seating positions;

at least one steerable wheel mechanically connected directly or indirectly to a steering control device operable by a driver to steer the amphibian, wherein the at least one steerable wheel and the steering control device operable by the driver remain connected throughout use in both the land and the marine modes such that there is no steering disconnect required during a change between land and marine modes; and

wherein the steering control device operable by the driver is moveable between at least two distinct laterally spaced positions across a width or beam of the amphibian including between a first position and a third position or between a second position and a third position, where the first position is laterally aligned with the first seating position, the second position is laterally aligned with the second seating position, and the third position is laterally aligned with the third seating position.

21. The amphibian of claim **20**, wherein the steering control device includes handlebars.

22. An amphibian operable on land and on water, comprising:

a planing hull;

at least one retractable wheel, the at least one retractable wheel being retractable from a protracted land engaging position when the amphibian is operated on land to a retracted position when the amphibian is operated on water;

at least one prime mover for providing power to drive, directly or indirectly, the amphibian on either one or both of land and water;

at least one steerable wheel mechanically connected directly or indirectly to a steering control device operable by a driver to steer the amphibian, wherein the at least one steerable wheel and the steering control device operable by the driver remain connected throughout use in both the land and the marine modes such that there is no steering disconnect required during a change between land and marine modes, and wherein the steering control device is positioned to be grasped by the driver while operating the amphibian; and wherein the steering control device operable by the driver is slidably moveable between at least two distinct laterally spaced positions across a width or beam of the amphibian, wherein the at least two distinct laterally spaced positions are substantially offset from one another during operation of the steering control device and comprise i) a center driving position of the amphibian and a right hand driving position of the amphibian, or ii) the center driving position of the amphibian and a left hand driving position of the amphibian.

* * * * *